MIDI 1.0

THE COMPLETE MIDI STANDARD

Musical Instrument Digital Interface · 1983 · studio-wall reference

All bytes in hex · **n** = channel 0–F (ch 1–16)

LOW LEVEL · THE WIRE

31 250	8-N-1	320 µs	3 125
bit/s, ±1 % tolerance	1 start + 8 data + 1 stop = 10 bits	per byte (10 ÷ 31 250 s)	bytes/s per cable

Asynchronous serial: no shared clock, every byte is framed by its own start and stop bit. Each cable is one-way, so **IN** and **OUT** run independently: one byte in and one byte out every 320 µs. Bit period 1/31 250 = 32 µs.

5 mA current loop on a 5-pin DIN (pins 4 and 5 carry the signal, pin 2 is the shield and is grounded at OUT only). Every IN is opto-isolated to avoid ground loops. THRU passes a copy of IN.

MESSAGE ANATOMY

1–3 bytes per message (SysEx: any length).

STATUS · 80–FF — bit 7 = 1 · type · channel

DATA · 00–7F — bit 7 = 0 · value 0–127

1	t	t	t	n	n	n	n
0	v	v	v	v	v	v	v

LSB + MSB data bytes = 14 bits (0–16 383): Pitch Bend, Song Position.

EVERY MESSAGE

STATUS	MESSAGE	DATA 1	DATA 2
CHANNEL VOICE			
8n	Note Off	key	release velocity
9n	Note On · vel 00 = Off	key	attack velocity
An	Polyphonic Key Pressure	key	pressure
Bn	Control Change	controller 00–77	value
Cn	Program Change	program 00–7F	–
Dn	Channel Pressure	pressure	–
En	Pitch Bend	LSB (00 = centre)	MSB (40 = centre)
CHANNEL MODE — sent as Control Change Bn 78–7F			
Bn 78	All Sound Off	00	
Bn 79	Reset All Controllers	00	
Bn 7A	Local Control	00 = off · 7F = on	
Bn 7B	All Notes Off	00	
Bn 7C	Omni Off · own channel	00	
Bn 7D	Omni On · all channels	00	
Bn 7E	Mono Mode On (Poly Off)	no. of channels (00 = per voice)	
Bn 7F	Poly Mode On (Mono Off)	00	
SYSTEM COMMON			
F0	System Exclusive start	manufacturer ID	data... until F7
F1	MTC Quarter Frame	0ttt dddd	–
F2	Song Position Pointer	LSB	MSB · in 16th notes
F3	Song / Sequence Select	song no.	–
F6	Tune Request	–	–
F7	End of Exclusive (EOX)	–	–
F4 F5	*undefined*		
SYSTEM REAL-TIME — single byte, no data			
F8	Timing Clock	24 per quarter note (24 PPQN)	
FA	Start	play from song position 0	
FB	Continue	play from current position	
FC	Stop		
FE	Active Sensing	optional · gaps ≤ 300 ms once used	
FF	System Reset	return to power-up state	
F9 FD	*undefined (F9 once proposed as 10 ms "tick")*		

CONTROL CHANGE · BN CC VV

DEC	HEX	CONTROLLER
COARSE / MSB — 0–127		
0	00	Bank Select
1	01	Modulation Wheel
2	02	Breath Controller
4	04	Foot Controller
5	05	Portamento Time
6	06	Data Entry (MSB)
7	07	Channel Volume (GM default 100)
8	08	Balance
10	0A	Pan (64 = centre)
11	0B	Expression
12	0C	Effect Control 1
13	0D	Effect Control 2
16–19	10–13	Gen. Purpose 1–4
3, 9, 14, 15, 20–31		*undefined*
FINE / LSB — adds precision, 0–16 383		
32–63	20–3F	LSB of CC 0–31 (n+32)
SWITCHES — 0–63 off · 64–127 on		
64	40	Sustain (Damper / Hold 1)
65	41	Portamento On/Off
66	42	Sostenuto
67	43	Soft Pedal
68	44	Legato Footswitch
69	45	Hold 2
SOUND CONTROLLERS		
70	46	SC 1 · Sound Variation
71	47	SC 2 · Timbre / Resonance
72	48	SC 3 · Release Time
73	49	SC 4 · Attack Time
74	4A	SC 5 · Brightness (Cutoff)
75	4B	SC 6 · Decay Time
76	4C	SC 7 · Vibrato Rate
77	4D	SC 8 · Vibrato Depth
78	4E	SC 9 · Vibrato Delay
79	*4F*	*SC 10 · undefined*

DEC	HEX	CONTROLLER
MORE		
80–83	50–53	Gen. Purpose 5–8
84	54	Portamento Control
88	58	High-Res Velocity Prefix
91	5B	Effects 1 · Reverb Send
92	5C	Effects 2 · Tremolo
93	5D	Effects 3 · Chorus Send
94	5E	Effects 4 · Celeste (Detune)
95	5F	Effects 5 · Phaser
96	60	Data Increment (+1)
97	61	Data Decrement (–1)
98	62	NRPN LSB
99	63	NRPN MSB
100	64	RPN LSB
101	65	RPN MSB
85–87, 89–90, 102–119		*undefined*
120–127	78–7F	Channel Mode (see left)

REGISTERED PARAMETERS (RPN 101/100)

00 00	Pitch Bend Sensitivity
00 01	Fine Tuning · 00 02 Coarse Tuning
00 03	Tuning Program · 00 04 Tuning Bank
00 05	Modulation Depth Range (GM2)
7F 7F	RPN Null: closes the parameter

Bend range ±12 on ch 1 (running status): `B0 65 00 64 00 06 0C` then close `65 7F 64 7F`

RECEIVER MODES

1	Omni On · Poly	all ch → polyphonic
2	Omni On · Mono	all ch → one voice
3	Omni Off · Poly	own ch → poly
4	Omni Off · Mono	one ch per voice

Mode messages 7C–7F also act as All Notes Off.

RUNNING STATUS

Consecutive channel messages with the same status byte need the status sent only once. Real-time bytes don't cancel it; System Common and SysEx do. Note On with velocity 00 = Note Off, which keeps running status going.

REAL-TIME CUTS IN. F8–FF may be inserted anywhere in the stream, even in the middle of another message. Receivers must handle them without losing the message they are parsing.

NOTE NUMBERS 0–127 · C-1 to G9



A0 = 21 · Middle C (C4) = 60 · A4 = 69 = 440 Hz · C8 = 108

Grand piano: 88 keys, notes 21–108. Yamaha names middle C "C3"; the note number is still 60.

MIDI CLOCK & PPQN

F8 is a steady pulse at **24 PPQN** (pulses per quarter note). After 24 clocks every device knows a quarter note has passed. Devices run faster internal clocks (48, 96, 480...) and convert. It syncs tempo (arpeggiators, BPM-based delays) but carries **no position**, so use it with Song Position Pointer or timecode. It can drift, so studios often add word clock or ADAT sync.

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MIDI 1.0

SysEx · GM / GS / XG · Extensions

Back: the addenda built on top of the 1983 core

SYSTEM EXCLUSIVE

`F0 id data… F7` data bytes are 7-bit (00–7F), any length.

`01–7C`	1-byte manufacturer ID
`00 xx xx`	3-byte extended ID
`7D`	Non-commercial / education
`7E`	Universal **Non-Real-Time**
`7F`	Universal **Real-Time**

Universal: `F0 7E|7F dev sub1 sub2 … F7` dev 7F = all devices.

COMMON MANUFACTURER IDS

`01`	Sequential	`41`	Roland
`04`	Moog	`42`	Korg
`07`	Kurzweil	`43`	Yamaha
`10`	Oberheim	`44`	Casio
`18`	E-mu	`47`	Akai
`40`	Kawai	`00 20 29`	Novation

READY-TO-SEND SYSEX

GM 1 System On	`F0 7E 7F 09 01 F7`
GM System Off	`F0 7E 7F 09 02 F7`
GM 2 System On	`F0 7E 7F 09 03 F7`
GS Reset (Roland)	`F0 41 10 42 12 40 00 7F 00 41 F7`
XG On (Yamaha)	`F0 43 10 4C 00 00 7E 00 F7`
Master Volume	`F0 7F 7F 04 01 ll mm F7`
Master Balance	`F0 7F 7F 04 02 ll mm F7`
Identity Request	`F0 7E 7F 06 01 F7`
Identity Reply	`F0 7E dev 06 02 mfr fam fam mod mod ver×4 F7`
MTC Full Frame	`F0 7F 7F 01 01 hr mn sc fr F7`
MMC Play / Stop	`F0 7F 7F 06 02 F7` / `… 06 01 F7`

Roland checksum: add the address and data bytes, then checksum = (128 – sum mod 128) mod 128. GS Reset: 40+00+7F+00 = BF; BF mod 80 = 3F; 80 – 3F = **41**.

UNIVERSAL NON-RT · 7E SUB-IDS

`01–03`	Sample Dump: header · packet · request
`04`	MIDI Time Code cueing
`05`	Sample Dump extensions
`06`	General Info (identity)
`07`	File Dump
`08`	MIDI Tuning Standard
`09`	General MIDI on/off
`7B–7F`	EOF · Wait · Cancel · NAK · ACK

UNIVERSAL RT · 7F SUB-IDS

`01`	MIDI Time Code (full frame)
`02`	MIDI Show Control
`03`	Notation information
`04`	Device Control (master vol/bal)
`05`	Real-time MTC cueing
`06`	MMC commands
`07`	MMC responses
`08`	MIDI Tuning (real-time)

MMC · F0 7F dev 06 cc F7

`01`	Stop	`07`	Record Exit
`02`	Play	`08`	Record Pause
`03`	Deferred Play	`09`	Pause
`04`	Fast Forward	`0A`	Eject
`05`	Rewind	`0B`	Chase
`06`	Record Strobe	`0D`	MMC Reset

Locate: `F0 7F dev 06 44 06 01 hr mn sc fr sf F7`

GM · GS · XG · GM2

GENERAL MIDI 1 · 1991

MMA/JMSC baseline. 128 programs in 16 families; drums on **channel 10** (keys 35–81); ≥ 24 voices; 16 channels. Must respond to CC 1, 7, 10, 11, 64, 121, 123, RPN 0–2, bend and channel pressure.

GS · ROLAND 1991

GM superset (SC-55). Variation tones via **CC 0 / CC 32**. Drum kits by Program Change on ch 10: Standard 1, Room 9, Power 17, Electronic 25, TR-808 26, Jazz 33, Brush 41, Orchestra 49, SFX 57. NRPN edits for filter, envelope and vibrato.

XG · YAMAHA 1994

GM superset (MU80). Bank MSB: **0** normal (LSB = variation), **64** SFX voices, **126** SFX kits, **127** drum kits, so any channel can play drums. ≥ 32 voices; reverb, chorus and variation effects.

GENERAL MIDI 2 · 1999

Standardises GS/XG ideas. Bank MSB **79h** = melodic, **78h** = rhythm; LSB picks the variation. ≥ 32 voices, two drum channels (10, 11), more CCs and RPN 05, plus master tuning and effect SysEx.

GM PROGRAM FAMILIES (1-based; send value – 1)

1–8	Piano	65–72	Reed
9–16	Chromatic Perc.	73–80	Pipe
17–24	Organ	81–88	Synth Lead
25–32	Guitar	89–96	Synth Pad
33–40	Bass	97–104	Synth Effects
41–48	Strings	105–112	Ethnic
49–56	Ensemble	113–120	Percussive
57–64	Brass	121–128	Sound Effects

FILE FORMATS

.MID is the Standard MIDI File (**SMF**): type 0 = one track, 1 = parallel tracks, 2 = independent patterns. **.KAR** is SMF with karaoke lyrics. Both usually target GM, GS, XG or GM2 so they play anywhere. Professionals use DAW project formats instead, which also store setup, audio, effects and takes. Ways to ship custom sounds without rendering audio: **DLS, SF2, RMID, XMF, XMI, SP-MIDI**.

GM DRUM MAP · CHANNEL 10

35	Acoustic Bass Dr.	47	Low-Mid Tom	59	Ride Cymbal 2	71	Short Whistle
36	Bass Drum 1	48	Hi-Mid Tom	60	Hi Bongo	72	Long Whistle
37	Side Stick	49	Crash Cymbal 1	61	Low Bongo	73	Short Guiro
38	Acoustic Snare	50	High Tom	62	Mute Hi Conga	74	Long Guiro
39	Hand Clap	51	Ride Cymbal 1	63	Open Hi Conga	75	Claves
40	Electric Snare	52	Chinese Cymbal	64	Low Conga	76	Hi Wood Block
41	Low Floor Tom	53	Ride Bell	65	High Timbale	77	Low Wood Block
42	Closed Hi-Hat	54	Tambourine	66	Low Timbale	78	Mute Cuica
43	High Floor Tom	55	Splash Cymbal	67	High Agogo	79	Open Cuica
44	Pedal Hi-Hat	56	Cowbell	68	Low Agogo	80	Mute Triangle
45	Low Tom	57	Crash Cymbal 2	69	Cabasa	81	Open Triangle
46	Open Hi-Hat	58	Vibraslap	70	Maracas		

MIDI EXTENSIONS

Sample Dump Standard (SDS)
Moves samples between devices as 120-byte SysEx packets with ACK/NAK/Wait handshaking (7E 01–03).

MIDI Time Code (MTC)
SMPTE time (hh:mm:ss:ff at 24, 25, 29.97 drop or 30 fps) carried as 8 F1 quarter frames per 2 frames, plus SysEx full frames for jumps. Wraps after 24 h. Give it its own port on a busy bus to avoid jitter.

MIDI Machine Control (MMC)
Transport for recorders and DAWs: play, stop, FF/rewind, pause, record, shuttle, locate/goto, cueing, record-ready. SysEx (7F … 06).

MIDI Show Control (MSC)
Real-time cues for lighting, sound, machinery and pyro in live and canned shows. Each cue in the controller's list maps to an MSC message; many devices now run it over Ethernet. SysEx (7F … 02).

MIDI Tuning Standard (MTS)
A tuning for each note: bulk dump request, bulk dump, and real-time single-note change. `note xx yy`: xx yy split a semitone into 16 384 steps (≈ 0.0061 cent).

MIDI Visual Control (MVC)
Lets MIDI drive video and visual presentation gear, using SysEx and ordinary channel messages.

3D Sound Control
RPNs for azimuth, elevation, distance and more, so synths can place sounds in 3D. All values 14-bit.

MIDI 2.0 · 2020
Universal MIDI Packet (32-bit words), 16 groups × 16 channels, 16-bit velocity, 32-bit controllers, per-note pitch and controllers, MIDI-CI negotiation and Property Exchange. Falls back to 1.0.

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